

# Madhu Lokesh

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## EDUCATION

### Old Dominion University

Master of Science, Computer Science (GPA: 3.94/4.0)

- **Coursework:** Machine Learning, Data Analytics and Big Data, Advanced Databases, Web Programming

Jan 2025 - Present

Norfolk, VA

## WORK EXPERIENCE

### Old Dominion University, Department of Computer Science | Graduate Research Assistant (GRA-Project Assistant)

Jan 2026 - Present

Advisor: Prof. Ravi Mukkamala

Paper: When Will the Next Attack Start? Calibrated Time-Window Prediction Using Multi-Agent LLMs (Submitted to IDSS 2026)

- Designed and implemented a production-quality Python pipeline using TensorFlow and a three-agent LLM system (Gemini-2.5-Pro and Gemini-2.5-Flash via Google Vertex AI) for cyberattack time-window prediction, leveraging natural language processing techniques, structured prompt engineering, agentic workflows with role-based coordination, and JSON-based inter-agent communication
- Designed and optimized an end-to-end RAG pipeline, incorporating advanced feature engineering, including data ingestion, chunking, embedding generation (Vertex AI gemini-embedding-001, 3,072 dimensions), vector storage in ChromaDB, and retrieval via cosine similarity over approximately 1,638 indexed historical traffic patterns with k=7 nearest-neighbor post-processing
- Built a cross-feature correlation engine and context builder with NumPy and Pandas that performs data preprocessing and computes Pearson correlations at lags -5 to +5, adaptive rolling statistics across multiple temporal scales, and first-order differences, expanding raw features to approximately 1,600 columns
- Implemented a confidence-aware peer-review agent to mitigate hallucinations by validating predictions against four weighted criteria (feature consistency, timing plausibility, confidence calibration, historical similarity), triggering targeted retries with corrective feedback when utility fell below 0.5
- Evaluated and benchmarked system performance across 50 trials on two datasets (CIC-IDS-2017 and a zero-day logistics-network corpus), achieving a Localization Score of 0.950 with 98% coverage and documenting improvements over single-agent baselines that dropped to 36% coverage on the harder dataset

### Cognizant Technology Solutions | Junior Software Engineer

Mar 2023 - Dec 2024

- Integrated RESTful APIs with MongoDB to build end-to-end data ingestion and retrieval pipelines for Sanofi, a global health services client, contributing to a 65% improvement in user satisfaction scores
- Designed interactive user interface enhancements and data dashboards with React, Chart.js, and jQuery that surfaced pharmaceutical sales insights, replacing manual reporting and leveraging data visualization techniques to improve stakeholder data visibility
- Built reusable React component libraries with TypeScript, enforcing strict type contracts that reduced runtime errors by 20% and accelerated feature delivery across teams
- Applied code-splitting and lazy loading to React bundles, cutting initial page load time by 40% and lowering bounce rates by 25%
- Configured Jenkins CI/CD pipelines for automated build, test, and deployment workflows, collaborating with QA engineers, product managers, and cross-functional stakeholders across Engineering and IT in an Agile environment

### Software Engineer Intern

Jun 2022 - Dec 2022

- Participated in the full SaaS development lifecycle within Agile/Scrum ceremonies including sprint planning, backlog refinement, retrospectives, and stakeholder demos, delivering weekly progress updates and documented findings
- Developed reusable UI components in Angular and React for a shared component library, improving development velocity by 30% on subsequent projects
- Integrated MySQL data sources into structured RESTful endpoints and conducted cross-browser compatibility testing, increasing user retention by 15%

## ACADEMIC PROJECTS

### Student Advising Portal | Old Dominion University

- Built and deployed a full-stack portal using React, Node.js, and MySQL where students select courses and advisors approve or reject selections across live frontend, backend, and database servers
- Implemented security hardening including reCAPTCHA, XSS protection, clickjacking defenses, two-factor authentication, and password policy enforcement

### Secondhand Car Price Prediction | Old Dominion University

- Performed exploratory data analysis with Python using Pandas and NumPy on used car sales data, conducting univariate, bivariate, and multivariate analysis to assess feature relationships and identify data preprocessing needs
- Engineered features and trained, evaluated, and benchmarked Linear Regression, K-Nearest Neighbors, and Random Forest models with scikit-learn and TensorFlow, incorporating natural language processing (NLP) techniques for text-based attributes and achieving an  $R^2$  score of 0.993 after systematic feature engineering and hyperparameter tuning; documented findings and recommendations

## SKILLS

- **Languages:** Python 3.11+, JavaScript, TypeScript, Java, SQL, HTML5, CSS3
- **AI/ML:** Large Language Models (LLMs), RAG Pipelines, Prompt Engineering, Agentic Workflows, Multi-Agent Systems, Vector Databases (ChromaDB), Embeddings, Hallucination Mitigation, Google Vertex AI, Scikit-learn, PyTorch, TensorFlow, Deep Learning, Natural Language Processing, Feature Engineering, Data Preprocessing
- **Frameworks & Tools:** React, Angular, Node.js, FastAPI, Django, REST APIs, Git, Docker, Linux/WSL, Jenkins CI/CD, Google Colab
- **Data:** Pandas, NumPy, MySQL, MongoDB, PostgreSQL, Apache Spark, Power BI, Looker, Data Visualization
- **Concepts:** Agile/Scrum, API Integration, Performance Benchmarking, Responsive Design, System Testing, Technical Documentation, Cloud Platforms (GCP)